Appendix A: Deriving the Bliss-Point Maximization Model

Solving for Q in Equation 2 yields:

$$Q = Q^{+} \pm \sqrt{\frac{\frac{U(H^{*},Q^{+})}{U(H,Q)} - 1 - \eta(H^{+} - H)^{2}}{\psi}}. \quad \text{(A1)}$$

At the choice point, the slope of the budget constraint (Equation 1) is equal to the first derivative of Equation A1 with respect to H,

$$\frac{dQ_N}{dH_N} = -(\tau + N/w)^{-1} = \frac{dQ}{dH} = -\frac{\eta|H^{*} - H|}{\sqrt{\psi\left[\frac{U(H^{*},Q^{+})}{U(H,Q)} - 1 - \eta(H^{*} - H)^{2}\right]}}. \quad \text{(A2)}$$

Equation 2 may be rearranged as:

$$\frac{U(H^{*},Q^{+})}{U(H,Q)} = 1 + \eta(H^{*} - H)^{2} + \psi(Q^{+} - Q)^{2}. \quad \text{(A3)}$$

Substituting Equation A3 for $U(H^{*},Q^{+})/U(H,Q)$ in Equation A2, setting $z = \psi/\eta$, $Q_N = Q$, $H_N = H$, and simplifying yields:

$$(\tau + N/w)^{-1} = \left(\frac{1}{z}\right)\left(\frac{H^{*} - H_N}{Q^{+} - Q_N}\right). \quad \text{(A4)}$$

To express demand Q_N in terms of price N without H_N, H_N may be solved for in Equation 1,

$$H_N = H^{*} - Q_N(\tau + N/w). \quad \text{(A5)}$$

Substituting Equation A5 for H_N in Equation A4, solving for Q_N, and then simplifying yields Equation 3, the Bliss-Point Maximization Model (BPMM).

Appendix B: Bayesian Hierarchical Analysis of the Bliss-Point Maximization Model (BPMM) for progressive ratio schedules of reinforcement

In the present paper we opted to estimate the BPMM (variant of MMT, Sanabria et al., 2025) via Bayes' theorem in STAN (e.g., STAN, Stan Development Team, 2024). This allowed us to easily account for the hierarchical organization of the data (responses nested in schedules in animals), while leveraging MCMC methods to estimate the distribution of potential values each free parameter can take.[1] What follows is a description of how we implemented MMT within STAN.

Let response rates (responses/second) of completed FR requirements be distributed as follows:

$$Log\left(RspRate_{N|j,s}\right) \sim N\left(Log\left(B_{N|j,s}\right), \sigma_{B_{j,s}}\right)$$

(B1)

where Log indicates the natural log, $Log\left(RspRate_{N|j,s}\right)$ is normally distributed with scale/mean $Log\left(B_{N|j,s}\right)$ and scale/standard deviation $\sigma_{B_{j,s}}$ indexed by ratio requirement *N*, lever height *j*, and subject *s;* note that *j* and *s* are treated as categorical; the subscript *B* on $\sigma_{B_{j,s}}$ just indicates it's the standard deviation of behavioral output. We assume $\sigma_{j,s}$ is potentially a function of lever height and $B_{N|s}$ is equal to Equation 4 from the text as:

$$B_{N|j,s} = \frac{Q_{N|j,s} N_{j,s}}{q_{j,s} H^{*}_{j,s}} * \frac{1}{60} \tag{B2}$$

[1] For more conceptual details, see our prior implementations (Daniels et al., 2018; Cheung et al., 2012; Thrailkill & Daniels, 2024) along with texts on implementation of Bayesian Hierarchical Analysis (BHA) such as Kruschke (2014).

where $Q_{N|j,s}$ is demand at ratio requirement $N_{j,s}$, $q_{j,s}$ is the magnitude of reward (here q_s = 1), $H^*_{j,s}$ is the duration of the session in minutes (here set to the total number of minutes completing FR requirements rather than total session time), and we scale this by 1/60 because we expressed *RspRate* as responses/second. Then, $Q_{N|j,s}$ is equal to Equation 3 from the text as:

$$Q_{N|j,s} = \frac{z_{j,s}Q^+_{j,s}}{z_{j,s}+\left(\tau_{j,s}+\frac{N_{j,s}}{w_{j,s}q_{j,s}}\right)^2} \tag{B3}$$

where $z_{j,s}$ is the ratio of the scalars of demand (ψ) and leisure (η; see Appendix A), $Q^+_{j,s}$ is the expected demand given no work requirement; $\tau_{j,s}$ is the number of minutes to utilize a unit of the reinforcer, and $w_{j,s}$ is responses per minute during work. Note that because $Q^+_{j,s}$ cannot take values larger than a subject would consume during the session ($H^*_{j,s}$) we calculated $Q^+_{j,s}$ as:

$$Q^+_{j,s} = \alpha_{j,s}H^*_{j,s}\frac{z_{j,s}\left[\frac{1}{\tau_{j,s}+\frac{N_{j,s}}{w_{j,s}q_{j,s}}}\right]^2+1}{z_{j,s}\left[\frac{1}{\tau_{j,s}+\frac{N_{j,s}}{w_{j,s}q_{j,s}}}\right]+\tau_{j,s}} \tag{B4}$$

where parameters are as before; $\alpha_{j,s}$ indicates the proportion of which $Q^+_{j,s}$ is equal to the right-hand side of the equation, and for simplification we set $N_{j,s} = 0$ thus eliminating everything but $\tau_{j,s}$ in the denominator of $\frac{1}{\tau_{j,s}+\frac{N_{j,s}}{w_{j,s}q_{j,s}}}$ to indicate that bliss-point demand is equal to the amount of the reinforcer a subject would consume if there was no work requirement.

Equations B1–B4 represents the Bliss-Point Maximization Model (BPMM); interpretation of free parameters can be found in Table 1. To account for the effect of lever height, let $\rho_{j,s,k}$ represent *k* free parameters $z_{j,s}$, $w_{j,s}$, $\tau_{j,s}$,$\sigma_{B_{j,s}}$, and $\alpha_{j,s}$ on the log or log-odds ($\alpha_{j,s}$) scale:

$$\rho_{j,s,k} = \begin{cases} \beta_{j=1,s,k} & if\ j = 1 \\ \beta_{j=1,s,k} + \beta_{j=11,s,k} & if\ j = 11 \\ \beta_{j=1,s,k} + \beta_{j=11,s,k} + \beta_{j=18,s,k} & if\ j = 18 \end{cases} \tag{B5}$$

where $\beta_{j,s,k}$ represent the effect of lever height *j* and thus $\beta_{j=1,s,k}$ is the intercept or effect on the lowest lever height of 1 cm, $\beta_{j=11,s,k}$ is the effect of the middle lever height of 11 cm, and $\beta_{j=18,s,k}$ is the effect of the highest lever height of 18 cm. Note, all parameters were back transformed prior to plugging into Equations B2–B4; and $\alpha_{j,s}$ was estimated on the log-odds to keep it constrained between zero and 1 when back transformed, it cannot be interpreted as an odds or probability. All lever height effects ($\beta_{j=1,s,k}$, $\beta_{j=11,s,k}$, $\beta_{j=18,s,k}$) were given the same prior distributions:

$$\begin{aligned} \beta_{j,s,k} &\sim N(\mu_{j,k}, \sigma_{j,k}) \\ \mu_{j,k} &\sim N(0,10) \\ \sigma_{j,k} &\sim IG(1,1) \end{aligned} \tag{B6}$$

where each effect was assumed to be described by a normal distribution with mean $\mu_{j,k}$ and $\sigma_{j,k}$, which in turn were described by a normal distribution with a mean/scale of zero and shape or standard deviation of 10 and an inverse-gamma distribution with scale and shape parameters equal to 1. These prior distributions were chosen to provide little to no information on our expectations regarding the effect of lever heights on performance in the progressive ratio schedule of reinforcement while ensuring that variability was likely small given that data was expressed as responses/second.

Taken together, Equations B1–B6 are taken as the BHA implementation of the BPMM for progressive ratio schedules of reinforcement, which was then estimated with cmdstan via the R interface. Initial fits of the model suggested posterior distributions of $\tau_{j,s}$ were approaching a point estimate of zero. Such an outcome is both incoherent (due to motoric limitations it must take some finite amount of time to utilize a reinforcer) and problematic in that it de-stabilized estimation of the other parameters. To overcome this problem, we added a hard coded minimum value by exploring values of $\tau_{j,s}$ until we found a reasonable value consistent with post-reinforcement pauses in the present data. Note that, the variational inference algorithm (see Kucukelbir et al., 2017) in STAN was used to rapidly explore $\tau_{j,s}$. After identifying that $\tau_{j,s}$ = 0.15 min/reinforcer struck a balance between fit and stable parameter estimates we generated estimates of our posterior distributions.

To get these estimates, we ran four chains with a warm-up of 5,000 draws followed by sampling 5,000 draws giving a total of 20,000 post warm-up draws to characterize each posterior distribution; acceptance criterion and treedepth of the MCMC sampler were 0.99 and 30, respectively. All posterior distributions met recommended convergence criteria: an effective sample size greater than or equal to 10% of total samples, a Monte Carlo Standard Error smaller than 10% of the standard deviation of the posterior distributions, a $\hat{R}$ no greater than 1.1, with no divergent transitions and E-BFMIs approximately equal to 0.3 but no less (see Stan Development Team, 2024, for a review of diagnostics of MCMC samplers). Inference was determined by the posterior distribution of the mean ($\mu_{j,k}$) of each *k* parameter. For code, input data, and output data, including visualization of fits to individuals see the associated Github repository: https://github.com/drcwadaniels/APSD_LeverHeight_BHA.

Appendix C: Posterior distribution moments and relationships

Table C1 summarizes the distributional moments of the posterior distribution of differences for each parameter. Inspection of this table confirms that only *w* and *z* changed with the increase from 1 to 11 cm and then plateaued. All other parameters (Q^+, τ, and σ), even when their means are different from zero, have 95 % credible intervals that contain zero. Note, however, that because of difficulties in estimating τ its insensitivity to manipulation should not be over interpreted; indeed, table C2 confirms τ did not appreciably change from our hard coding of τ = 0.15 min/reinforcer.

Table C1 Mean and 95% credible interval of the posterior distribution of differences for each parameter across lever conditions

	1→11 cm		11→18 cm	
Parameter	Mean	95% Credible Interval	Mean	95% Credible Interval
w	1.28	0.39, 2.69	0.11	-0.31, 0.49
z	-1.75	-4.39, -0.36	-0.01	-0.66, 0.69
Q^+	8.57	-5.85, 23.89	3.4	-11.46, 21.08
τ	-8.21	-23.69, 5.77	-3.54	-21.13, 11.11
σ	-0.09	-0.29, 0.11	0.05	-0.14, 0.26

Note. Differences are on the Natural Log Scale. Q^+ is represented as the difference in the log-odds and thus proportion of which Q^+ equals the theoretical limit of demand (see Appendix B, Eq. B4); σ is variability around log-normally distributed responses (see Appendix B, Eq. B1). The 95% Credible Interval is defined by the Lower,Upper percentiles

Table C2 summarizes the mean and inter-quartile range of distribution of the means of each parameter across animals. Inspection of this table reveals that as the lever height increased, animals went from nine to 42 responses/min when working; and the ratio of work to leisure decreased from 1.79 to 0.22. Insofar as *z* is an exchange rate of demand for leisure it indicates that the exchange rate decreased in favor of leisure, suggesting that as the lever height increased, so did the value of leisure. Additionally, total demand did not change across lever height conditions remaining around 191–197 reinforcers per session. Likewise, variability in responding remained relatively constant.

Table C2 Mean and inter-quartile range (IQR) of the means of the posterior distributions of each parameter across animals in each condition

	1 cm		11 cm		18 cm	
Parameter	Mean	IQR	Mean	IQR	Mean	IQR
w	9.42	7.81, 11.37	41.89	22.79, 79.15	42.49	23.54, 96.36
z	1.79	1.23, 3.09	0.22	0.15, 0.56	0.18	0.10, 0.68
Q^+	191.44	168.52, 197.55	196.74	163.80, 214.20	192.81	159.56, 208.71
τ	0.15	0.15, 0.15	0.15	0.15, 0.15	0.15	0.15, 0.15
σ	0.47	0.42, 0.54	0.45	0.41, 0.48	0.49	0.43, 0.51

Note. Estimates here are on the natural rather than natural-log scale; σ is variability around log-normally distributed responses (see Appendix B, Eq. B1). The IQR is defined by the first and third quartiles

We also inspected whether subject-specific estimates summarized in Table C2 correlated within each condition when on the natural log scale. In the 1-cm lever condition, we found a significant correlation ($r(6) = 0.76$, $p = 0.027$, $p_{FDR} = 0.214$) between Q^+ and z. In the 11-cm lever condition we found a significant correlation between w and z ($r(6) = -0.85$, $p = 0.006$, $p_{FDR} = 0.10$). In the 18-cm lever condition, we found a significant correlation of similar magnitude between w and z ($r(6) = -0.91$, $p = 0.001$, $p_{FDR} = 0.048$). This pattern of correlation suggests a positive relationship between total demand and the exchange rate for leisure under the 1-cm lever condition that then disappears once the lever is raised to 11 and 18 cm. However, once the lever is raised to 11 and 18 cm, we see a strong, negative correlation between work (w) and the exchange rate for leisure (z), suggesting the more an animal works the more they value leisure. While interesting, we caution that correlations at low sample sizes are volatile (De Winter et al., 2016) and, in the present study, only the relationship between w and z when the lever height is 18 cm survives a modest correction for the number of tests conducted by controlling the false discovery rate (FDR; lowest p_{FDR} = 0.048). It is unclear why this relationship should hold under the 18-cm and not the 11-cm condition. Additionally, even if the relationship between any parameters is true, r = 0.76 to r = -0.91 indicates an r^2 = 0.58 to r^2 = 0.83, thus leaving a substantial amount of variation in parameters to be explained by some exogenous variable rather than a different free parameter. Nonetheless, future work may aim to replicate and understand how the relationship between parameters changes across lever heights.